

AZHL

Order Cancellation & Customer Refund

Backend API & Business Logic Specification

Customer | Provider | Marketplace | Admin Monitoring

CORE RULE

If a paid order is cancelled before financial completion by the Customer, Provider, or Marketplace, the full paid order value must be refunded to the Customer. No Azhl platform fee is deducted on a cancelled order.

Example: Customer paid 50 SAR -> Order cancelled -> Customer refund = 50 SAR

1. Business Rule

- Cancellation can be initiated by the Customer, Provider, or Marketplace.
- Cancellation is allowed only while the order is still eligible for cancellation and has not been financially settled as completed.
- If payment was successfully collected, the full captured order amount must be refunded to the same Customer payment source whenever the payment gateway supports it.
- Azhl fee is not charged on a cancelled order because the service/order was not completed.
- Any provider/marketplace pending amount related to the cancelled order must be removed immediately from pending earnings.
- A completed/refunded order must not be processed twice.

2. 50 SAR Example

CUSTOMER PAYS 50 SAR	AZHL HOLDS 50 SAR	ORDER CANCELLED	REFUND 50 SAR
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Platform Fee = 0 SAR | Provider/Marketplace Earning = 0 SAR | Customer Refund = 50 SAR

3. Cancellation Actors

Actor	Allowed Action	Refund Recipient	Required Data
Customer	Cancel eligible order	Customer	Cancellation reason
Provider	Cancel eligible assigned order	Customer	Cancellation reason
Marketplace	Cancel eligible marketplace order	Customer	Cancellation reason

Important

Backend should infer cancelled_by from the authenticated account/role. The client should not be trusted to submit a different actor ID.

4. Order & Refund Statuses

Order Status

active / accepted / in_progress -> cancelled
 completed -> cancellation not allowed
 cancelled -> cannot cancel again

Refund Status

not_required Payment was never captured
 requested Refund record created
 processing Refund sent to payment gateway
 refunded Customer refund completed
 failed Gateway/manual refund failed and needs retry/review

5. Cancellation API

POST /api/v1/orders/{order_id}/cancel
 Authorization: Bearer {token}
 Content-Type: application/json

Request

```
{
  "reason": "Provider is unavailable and cannot complete this order."
}
```

- reason: required, string, recommended max 1000 characters.
- Backend validates that the authenticated Customer/Provider/Marketplace is actually related to the order.
- Backend validates that the order status is cancellable.
- Backend checks the payment status before deciding whether a refund is required.

6. Successful Cancellation - Refund Processing

```
{
  "success": true,
  "message": "Order cancelled successfully. Refund is being processed.",
  "data": {
    "order_id": 812,
    "order_no": "ORD-000812",
    "order_status": "cancelled",
    "cancelled_by": "provider",
    "cancellation_reason": "Provider is unavailable.",
    "cancelled_at": "2026-08-19T11:45:00+05:00",
    "payment": {
      "paid_amount": 50.00,
      "currency": "SAR"
    },
    "refund": {
      "amount": 50.00,
      "currency": "SAR",
      "status": "processing",
      "refund_reference": null
    }
  }
}
```

7. Successful Refund Response

```
{
  "success": true,
  "message": "Order cancelled and customer refunded successfully.",
  "data": {
    "order_id": 812,
    "order_status": "cancelled",
    "refund": {
      "amount": 50.00,
      "currency": "SAR",
      "status": "refunded",
      "refund_reference": "REF_9X81AB23",
      "refunded_at": "2026-08-19T11:46:12+05:00"
    }
  }
}
```

8. No Captured Payment Case

Scenario

If an order is cancelled before a successful payment/capture exists, cancellation still succeeds but no gateway refund is required.

```
{
  "order_status": "cancelled",
  "refund": {
    "status": "not_required",
    "amount": 0.00
  }
}
```

9. Wallet / Pending Balance Adjustment

Provider Order

Before cancellation:

Order Amount = 50 SAR

Provider Pending Amount includes 50 SAR

After cancellation:

Provider Pending Amount -= 50 SAR

Provider Available Balance += 0 SAR

Azhl Fee += 0 SAR

Customer Refund = 50 SAR

Marketplace Order

Before cancellation:

Marketplace pending amount includes the paid order amount

After cancellation:

Remove cancelled order amount from Marketplace pending

Do not create marketplace earning

Refund the full paid amount to Customer

No Credit Transaction

A cancelled order must not create a Provider/Marketplace CREDIT wallet transaction because no earning was finalized. The cancellation/refund should remain visible in order/payment/refund history.

10. Refund Record

- Create one refund record per refund attempt / logical refund according to the payment integration design.
- Store the original payment reference/charge ID.
- Store the payment gateway refund reference once returned.
- Store refund amount and currency.
- Store requested_at, processed_at/refunded_at, failed_at and failure_reason as applicable.
- Store who caused the cancellation through cancelled_by on the order/cancellation record.

Recommended refunds Table

```

id
refund_no
order_id
payment_id
customer_id
amount
currency
status
gateway
gateway_refund_id
failure_reason
requested_at
refunded_at
failed_at
created_at
updated_at

```

11. Recommended Order Cancellation Fields

```

order_status
cancelled_by_type    customer | provider | marketplace
cancelled_by_id
cancellation_reason
cancelled_at
refund_status
refund_id

```

12. Payment Gateway Refund Logic

LOCK ORDER	VALIDATE CANCELLATION	CREATE REFUND	CALL PAYMENT GATEWAY	UPDATE REFUND STATUS
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- Use the original successful payment/charge reference for the refund.
- Refund amount must equal the amount actually captured for the order when the business rule is full refund.
- Do not mark refund as refunded before the gateway confirms success.
- Persist gateway response/reference needed for reconciliation.
- If gateway call fails, keep the order cancelled but mark refund as failed/processing according to the failure type and expose it for retry/admin review.

13. Failed Refund Handling

```
{
  "success": true,
  "message": "Order cancelled, but refund requires review.",
  "data": {
    "order_status": "cancelled",
    "refund": {
      "amount": 50.00,
      "status": "failed",
      "failure_reason": "Payment gateway refund request failed."
    }
  }
}
```

Critical

Do not reactivate the order just because the refund API fails. Cancellation and refund settlement are separate states. Failed refunds should be retried or handled by Admin without creating duplicate refunds.

14. Admin Refund Monitoring

```
GET /api/v1/admin/refunds?status=failed&page=1
GET /api/v1/admin/refunds?status=processing&page=1
GET /api/v1/admin/refunds?status=refunded&page=1
```

- Admin should see Order No., Customer, Cancelled By, Cancel Reason, Paid Amount, Refund Amount, Refund Status, Gateway Reference and timestamps.
- Admin should be able to retry failed refunds only through a protected/idempotent backend action.
- Manual completion should require a reference/note if the refund is settled outside the automatic payment gateway flow.

15. Business Validation Errors

Completed Order Cannot Be Cancelled

```
HTTP 422
{
  "success": false,
  "message": "Completed orders cannot be cancelled.",
  "data": null
}
```

Unauthorized Actor

```
HTTP 403
{
  "success": false,
  "message": "You are not authorized to cancel this order."
}
```

Already Cancelled

```
HTTP 422
{
  "success": false,
  "message": "This order has already been cancelled."
}
```

16. Database Transaction / Locking

BEGIN TRANSACTION

1. Lock order row
2. Validate actor + order state
3. Mark order cancelled once
4. Remove order amount from Provider/Marketplace pending balance
5. Create cancellation/refund record
6. Commit cancellation state

Refund gateway processing executes with idempotency protection and updates refund status without creating a duplicate refund.

17. Idempotency Rules

- The same order cannot be cancelled twice.
- The same captured payment cannot be fully refunded twice.
- A retry must reuse/recognize the existing refund record instead of creating another 50 SAR refund.
- Wallet pending balance must be reduced only once.
- Gateway webhooks and API responses must be safe if delivered more than once.

18. Final End-to-End Flow

Customer / Provider / Marketplace cancels order
Backend validates actor and cancellable order status
Order status -> CANCELLED
Remove order amount from Provider / Marketplace pending
Check successful captured payment
Create full refund for Customer
Refund amount = original paid order value
Azhl fee = 0 Provider/Marketplace earning = 0
Gateway confirms refund -> REFUNDED

19. Backend Acceptance Checklist

- ✓ Customer can cancel own eligible order.
- ✓ Assigned Provider can cancel eligible order.
- ✓ Related Marketplace can cancel eligible order.
- ✓ Full captured amount is refunded to Customer.
- ✓ No Azhl fee is deducted on cancelled order.
- ✓ Provider/Marketplace pending amount is removed once.
- ✓ No credit earning is created for cancelled order.
- ✓ Refund has requested/processing/refunded/failed state.
- ✓ Completed order cannot be cancelled through normal cancellation flow.
- ✓ Duplicate cancellation/refund is prevented.
- ✓ Admin can monitor failed/processing refunds.
- ✓ All timestamps and gateway references are stored for audit.

Final Rule

PAID + CANCELLED BEFORE COMPLETION = FULL CUSTOMER REFUND. For a 50 SAR paid order: Customer gets 50 SAR back, Azhl fee is 0 SAR, and Provider/Marketplace earning is 0 SAR.